A FINITE-TYPE QUASI-SEPARATED ALGEBRAIC STACK EMBEDS CLOSED INTO A FINITELY PRESENTED STACK OVER A QCQS BASE

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Editorial note. The stack morphism is only quasi-separated, not assumed separated. The separated case used in the preparedGabber–Chow theorem is supporting context, not this general statement.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Algebraic stacks and smooth groupoid presentations, finite-typeqcqs approximation of algebraic spaces, finite-presentation descent of morphisms and diagrams, quotient stacks and local criteria for closed immersions.

Source reference. Original proposition 07SU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9914–9914. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0CP5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15915–15915. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0230: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2725–2725. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0CPB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15921–15921. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0CPD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15923–15923. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0CPC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15922–15922. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:stacks-limits.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0CPF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15925–15925. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0CPA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15920–15920. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original theorem 04TK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6258–6258. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 046Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5481–5481. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04ZN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6471–6471. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0CP9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15919–15919. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0503: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6486–6486. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *Let $f : \mathcal{X} \to Y$ be a morphism from an algebraic stack to an algebraic space. Assume:*

(1) *f is of finite type and quasi-separated,*
(2) *Y is quasi-compact and quasi-separated.*

Then there exists a morphism of finite presentation $f' : \mathcal{X}' \to Y$ and a closed immersion $\mathcal{X} \to \mathcal{X}'$ of algebraic stacks over Y.

Proof. Write $Y = \lim_{i \in I} Y_i$ as a limit of algebraic spaces over a directed set I with affine transition morphisms and with Y_i Noetherian, see Limits of Spaces, Proposition 07SU. We will use the material from Limits of Spaces, Section 0CP5.

Choose a presentation $\mathcal{X} = [U/R]$. Denote (U, R, s, t, c, e, i) the corresponding groupoid in algebraic spaces over Y. We may and do assume U is affine. Then U, R, $R \times_{s,U,t} R$ are quasi-separated algebraic spaces of finite type over Y. We have two morpisms $s, t : R \to U$, three morphisms $c : R \times_{s,U,t} R \to R$, $\mathrm{pr}_1 : R \times_{s,U,t} R \to R$,

$\mathrm{pr}_2 : R \times_{s,U,t} R \to R$, a morphism $e : U \to R$, and finally a morphism $i : R \to R$. These morphisms satisfy a list of axioms which are detailed in Groupoids, Section 0230.

According to Limits of Spaces, Remark 0CPB we can find an $i_0 \in I$ and inverse systems

(1) $(U_i)_{i \geq i_0}$,
(2) $(R_i)_{i \geq i_0}$,
(3) $(T_i)_{i \geq i_0}$

over $(Y_i)_{i \geq i_0}$ such that $U = \lim_{i \geq i_0} U_i$, $R = \lim_{i \geq i_0} R_i$, and $R \times_{s,U,t} R = \lim_{i \geq i_0} T_i$ and such that there exist morphisms of systems

(1) $(s_i)_{i \geq i_0} : (R_i)_{i \geq i_0} \to (U_i)_{i \geq i_0}$,
(2) $(t_i)_{i \geq i_0} : (R_i)_{i \geq i_0} \to (U_i)_{i \geq i_0}$,
(3) $(c_i)_{i \geq i_0} : (T_i)_{i \geq i_0} \to (R_i)_{i \geq i_0}$,
(4) $(p_i)_{i \geq i_0} : (T_i)_{i \geq i_0} \to (R_i)_{i \geq i_0}$,
(5) $(q_i)_{i \geq i_0} : (T_i)_{i \geq i_0} \to (R_i)_{i \geq i_0}$,
(6) $(e_i)_{i \geq i_0} : (U_i)_{i \geq i_0} \to (R_i)_{i \geq i_0}$,
(7) $(i_i)_{i \geq i_0} : (R_i)_{i \geq i_0} \to (R_i)_{i \geq i_0}$

with $s = \lim_{i \geq i_0} s_i$, $t = \lim_{i \geq i_0} t_i$, $c = \lim_{i \geq i_0} c_i$, $\mathrm{pr}_1 = \lim_{i \geq i_0} p_i$, $\mathrm{pr}_2 = \lim_{i \geq i_0} q_i$, $e = \lim_{i \geq i_0} e_i$, and $i = \lim_{i \geq i_0} i_i$. By Limits of Spaces, Lemma 0CPD we see that we may assume that s_i and t_i are smooth (this may require increasing i_0). By Limits of Spaces, Lemma 0CPC we may assume that the maps $R \to U \times_{U_i, s_i} R_i$ given by s and $R \to R_i$ and $R \to U \times_{U_i, t_i} R_i$ given by t and $R \to R_i$ are isomorphisms for all $i \geq i_0$. By Limits of Spaces, Lemma 0CPF we see that we may assume that the diagrams

$$\begin{array}{ccc} T_i & \xrightarrow[q_i]{} & R_i \\ {\scriptstyle p_i}\downarrow & & \downarrow{\scriptstyle t_i} \\ R_i & \xrightarrow{s_i} & U_i \end{array}$$

are cartesian. The uniqueness of Limits of Spaces, Lemma 0CPA then guarantees that for a sufficiently large i the relations between the morphisms s, t, c, e, i mentioned above are satisfied by s_i, t_i, c_i, e_i, i_i. Fix such an i.

It follows that $(U_i, R_i, s_i, t_i, c_i, e_i, i_i)$ is a smooth groupoid in algebraic spaces over Y_i. Hence $\mathcal{X}_i = [U_i/R_i]$ is an algebraic stack (Algebraic Stacks, Theorem 04TK). The morphism of groupoids

$$(U, R, s, t, c, e, i) \to (U_i, R_i, s_i, t_i, c_i, e_i, i_i)$$

over $Y \to Y_i$ determines a commutative diagram

$$\begin{array}{ccc} \mathcal{X} & \longrightarrow & \mathcal{X}_i \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Y_i \end{array}$$

(Groupoids in Spaces, Lemma 046Q). We claim that the morphism $\mathcal{X} \to Y \times_{Y_i} \mathcal{X}_i$ is a closed immersion. The claim finishes the proof because the algebraic stack $\mathcal{X}_i \to Y_i$ is of finite presentation by construction. To prove the claim, note that the left diagram

$$\begin{array}{ccc} U & \longrightarrow & U_i \\ \downarrow & & \downarrow \\ \mathcal{X} & \longrightarrow & \mathcal{X}_i \end{array} \qquad \begin{array}{ccc} U & \longrightarrow & Y \times_{Y_i} U_i \\ \downarrow & & \downarrow \\ \mathcal{X} & \longrightarrow & Y \times_{Y_i} \mathcal{X}_i \end{array}$$

is cartesian by Groupoids in Spaces, Lemma 04ZN and the results mentioned above. Hence the right commutative diagram is cartesian too. Then the desired result follows from the fact that $U \to Y \times_{Y_i} U_i$ is a closed immersion by construction of the inverse system (U_i) in Limits of Spaces, Lemma 0CP9, the fact that $Y \times_{Y_i} U_i \to Y \times_{Y_i} \mathcal{X}_i$ is smooth and surjective, and Properties of Stacks, Lemma 0503. $\square$